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Laser Assisted Radiative Recombination

Project Report

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Abstract

This project investigates Laser-Assisted Radiative Recombination (LARR) of free electrons with hydrogenic ions under intense, few-cycle laser pulses. Building on the theoretical framework of Bivona et al. (2006, 2007), we implemented computational simulations using both a full Quantum Mechanical approach based on the Coulomb-Volkov approximation and a simplified Semiclassical model. We successfully replicated key spectral features from the literature and extended the analysis by performing detailed carrier-envelope phase (CEP) parameter sweeps for two and three cycle pulses. Our results show that the spectral cutoff positions and overall width are highly sensitive to the CEP in the few-cycle regime, confirming that the emission spectrum can be controlled by manipulating the CEP.

Contents

1	Introduction	3
1.1	Background	3
1.2	Motivation and Objectives	3
2	Theoretical Framework	5
2.1	Physical Picture	5
2.2	The Laser Field	5
2.3	Quantum Mechanical Model	6
2.3.1	Coulomb–Volkov Approximation	6
2.3.2	Double Differential Probability	7
2.4	Semiclassical Model	7
2.4.1	Two-Step Process	7
2.4.2	Classical Differential Probability	8
3	Computational and Numerical Implementation	9
3.1	Choice of Units and Physical Parameters	9
3.2	Time Grid and Pulse Construction	9
3.3	Evaluation of the Spatial Matrix Element	10
3.4	Quantum-Mechanical Probability	11
3.5	Semiclassical Probability and Root Finding	12
3.6	Parallelization and Performance Considerations	13
4	Results and Discussion	14
4.1	Validation Against Literature	14
4.2	CEP Parameter Sweeps	16
5	Limitations and Future Directions	18
5.1	Validity of the Coulomb–Volkov Approximation	18
5.2	Neglect of Electron–Electron and Plasma Effects	18
5.3	Numerical Constraints	19
5.4	Possible Extensions	19

6 Conclusion	20
Acknowledgments	23

Chapter 1

Introduction

1.1 Background

Radiative Recombination (RR) is a basic atomic process in which a free electron recombines with a positive ion and emits a photon. This process occurs in many physical systems, including astrophysical plasmas and laboratory environments. When the recombination takes place in the presence of an intense laser field, it is known as Laser-Assisted Radiative Recombination (LARR). In this case, the laser field can exchange energy with the electron during recombination, which leads to a broadening of the emitted photon spectrum compared to the field-free case.

Recent developments in laser technology have made it possible to generate few-cycle laser pulses, where the total pulse duration contains only a small number of optical cycles. In this regime, the Carrier-Envelope Phase (CEP), defined as the phase difference between the pulse envelope and the carrier wave, becomes an important parameter. Changing the CEP alters the shape of the electric field in time, which can strongly affect the motion of the electron and, as a result, the emitted radiation.

1.2 Motivation and Objectives

The main objectives of this project are:

1. To computationally reproduce laser-assisted radiative recombination spectra for hydrogen driven by few-cycle Ti:Sapphire laser pulses, using both quantum-mechanical and semiclassical approaches.
2. To study how the emitted photon spectrum depends on the carrier-envelope phase, with particular focus on the spectral cutoffs.
3. To extend earlier studies by performing detailed CEP scans from -90° to $+90^\circ$ and presenting the results in the form of heatmaps.

4. To understand the physical origin of the observed features using both quantum and classical pictures.

Through these objectives, this project aims to show that LARR in the few-cycle regime can be influenced by the carrier-envelope phase and to gain physical insight into how this control arises.

Chapter 2

Theoretical Framework

2.1 Physical Picture

Before discussing the mathematical formulation, it is useful to build some physical intuition for the process. Consider a free electron approaching a positive ion. In the absence of any external field, the electron recombines into a bound state and emits a photon with energy equal to the sum of the ionization potential and the kinetic energy of the incoming electron.

When an oscillating laser field is present, the situation changes. The electron no longer follows a simple straight-line trajectory but instead oscillates under the influence of the laser field. This laser-driven quiver motion modifies the electron's instantaneous momentum at the time of recombination. As a result, the emitted photon energy is no longer fixed, but spans a range of values. This range is bounded by the minimum and maximum values of the electron's instantaneous kinetic energy during the laser pulse, defining a classically allowed energy window.

For few-cycle laser pulses, the dynamics become particularly sensitive to the carrier-envelope phase (CEP). Since the electron experiences only a small number of oscillations, changes in the CEP significantly alter the temporal shape of the electric field. Different CEP values lead to different waveform asymmetries, which in turn produce different electron trajectories and recombination energies. This sensitivity provides a natural mechanism for controlling the emission spectrum in laser-assisted radiative recombination.

2.2 The Laser Field

We model the laser pulse as a linearly polarized wave with a $\sin^2$ envelope:

$$E(t) = \begin{cases} E_0 \sin^2\left(\frac{\pi t}{\tau}\right) \cos(\omega t + \varphi), & 0 \leq t \leq \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (2.1)$$

Here $\tau = n_c T$ is the total pulse duration with n_c optical cycles and period $T = 2\pi/\omega$, E_0 is the peak electric field amplitude, and φ is the carrier-envelope phase.

The corresponding vector potential is obtained by integration,

$$A(t) = -c \int_0^t E(t') dt', \quad (2.2)$$

and the momentum transferred to the electron by the laser field is

$$k_L(t) = \frac{A(t)}{c}. \quad (2.3)$$

The instantaneous electron momentum during the pulse is therefore

$$q(t) = q + k_L(t), \quad (2.4)$$

where q is the asymptotic electron momentum when the laser field is switched off.

2.3 Quantum Mechanical Model

2.3.1 Coulomb–Volkov Approximation

The quantum-mechanical treatment of laser-assisted radiative recombination involves calculating the transition amplitude for an electron initially in a continuum state to recombine into a bound state while emitting a photon. The interaction with the laser field is treated exactly, while the electron–ion interaction is included through the Coulomb–Volkov approximation.

In this approximation, the dressed continuum electron wavefunction is written as

$$\Psi_q^+(\mathbf{r}, t) = \exp \left[i[q + k_L(t)] \cdot \mathbf{r} - \frac{i}{2} \int_0^t [q + k_L(t')]^2 dt' \right] u_q^+(\mathbf{r}), \quad (2.5)$$

where $u_q^+(\mathbf{r})$ is the incoming Coulomb scattering state.

The spatial part $u_q^+(\mathbf{r})$ is given by

$$u_q^+(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} e^{\pi\nu/2} \Gamma(1 - i\nu) {}_1F_1(-i\nu, 1, i(qr - \mathbf{q} \cdot \mathbf{r})), \quad (2.6)$$

where $\nu = Z/q$, Γ is the Gamma function, and ${}_1F_1$ is the confluent hypergeometric function. This form accounts for the long-range Coulomb interaction between the electron and the ion.

The bound state is approximated by the unperturbed ground state of the hydrogenic atom with $(Z-1)$ charge.

$$\Psi_0(\mathbf{r}, t) = u_0(\mathbf{r}) e^{-iI_0 t}, \quad (2.7)$$

with binding energy $I_0 = -Z^2/2$.

2.3.2 Double Differential Probability

Using first-order perturbation theory for the electron–photon interaction, the double differential probability (DDP) for emitting a photon of energy ω_X at the end of the pulse is given by

$$\text{DDP}(\tau) = \frac{\omega_X^3}{(2\pi)^2 c^3} \left| \int_0^\tau dt' M_q(t') \exp \left[i(\omega_X + I_0)t' - \frac{i}{2} \int_0^{t'} [q + k_L(t'')]^2 dt'' \right] \right|^2. \quad (2.8)$$

The matrix element $M_q(t')$ contains the spatial overlap between the bound and continuum states and is defined as

$$M_q(t') = \langle u_0(\mathbf{r}) | \boldsymbol{\varepsilon}_X \cdot \mathbf{r} e^{i\mathbf{k}_X \cdot \mathbf{r}} e^{i[q + k_L(t')] \cdot \mathbf{r}} | u_q^+(\mathbf{r}) \rangle, \quad (2.9)$$

where $\boldsymbol{\varepsilon}_X$ and $\mathbf{k}_X$ are the polarization and wavevector of the emitted photon.

The coherent time integral implies that recombination amplitudes from different emission times interfere. As a result, the quantum spectrum exhibits oscillatory structures arising from constructive and destructive interference between different recombination pathways.

2.4 Semiclassical Model

The semiclassical model provides a simpler and more intuitive description by treating the electron motion classically while retaining a quantum-mechanical description of recombination.

2.4.1 Two-Step Process

Step 1: The electron propagates in the laser field with instantaneous momentum

$$q(t) = q + k_L(t), \quad (2.10)$$

and kinetic energy $\varepsilon(t) = [q + k_L(t)]^2/2$.

Step 2: At a recombination time t_r , the electron recombines instantaneously with the ion. Energy conservation then gives the emitted photon energy

$$\omega_X(t_r) = \frac{1}{2}[q + k_L(t_r)]^2 + |I_0|. \quad (2.11)$$

For a fixed photon energy ω_X , all times t_l satisfying $\omega_X = \omega_X(t_l)$ contribute to the

emission.

2.4.2 Classical Differential Probability

The classical differential probability (CDP) is obtained by summing incoherently over all contributing times:

$$\text{CDP}(\tau) = \sum_l \frac{\omega_X^3}{2\pi c^3} \frac{|M(t_l)|^2}{|E(t_l) \cdot [q + k_L(t_l)]|}. \quad (2.12)$$

The denominator is proportional to $|d\omega_X/dt|$ evaluated at $t = t_l$ and diverges when $E(t_l) \approx 0$. Physically, these divergences correspond to turning points of the electron motion, where the electron spends a longer time at a given energy. These points give rise to enhanced emission near the spectral cutoffs.

While the semiclassical model does not capture quantum interference effects, it accurately predicts the spectral envelope and cutoff positions and provides clear physical insight into the recombination process.

Chapter 3

Computational and Numerical Implementation

The theoretical expressions presented in the previous chapter involve multi-dimensional integrals over time and space and do not admit closed-form solutions for few-cycle laser pulses. Consequently, numerical implementation plays a central role in this project. This chapter describes the computational strategy adopted to evaluate both the quantum-mechanical and semiclassical recombination probabilities, along with the numerical challenges encountered and the choices made to balance accuracy and feasibility.

All simulations were implemented in `Python`, using `NumPy` for numerical operations and `SciPy` for special functions and quadrature routines. Additionally, `Flint` was used for hypergeometric calculations, which significantly reduced the simulation time.

3.1 Choice of Units and Physical Parameters

All calculations were carried out in atomic units, where $\hbar = m_e = e = 1$. This choice significantly simplifies the expressions for the laser–electron interaction and avoids repeated conversions between physical constants. Energies were converted to electron volts only at the post-processing stage for plotting and comparison with literature results.

The laser and atomic parameters were chosen to match those used in Refs. [1, 2], allowing direct validation of the numerical implementation.

3.2 Time Grid and Pulse Construction

The few-cycle laser pulse was discretized on a uniform time grid over the interval $[0, \tau]$, with $\tau = n_c T$ and $T = 2\pi/\omega$. The electric field was constructed using a $\sin^2$ envelope, and the vector potential was obtained by numerical integration.

A representative implementation is shown below.

```

def make_pulse(nc, phi_deg, n_t=4000):
    tau = nc * (2*np.pi/omega_au)
    t = np.linspace(0, tau, n_t, endpoint=False)
    phi = np.deg2rad(phi_deg)

    env = np.sin(np.pi * t / tau)**2
    E_t = E0_au * env * np.cos(omega_au * t + phi)

    dt = t[1] - t[0]
    A_t = -c_au * np.cumsum(E_t) * dt

    kL_t = A_t / c_au
    q_t = q0 + kL_t
    omega_t = 0.5 * q_t**2 + 0.5 * Z**2

    return t, E_t, kL_t, q_t, omega_t, tau

```

The instantaneous electron momentum $q(t)$ and energy $\omega_X(t)$ were precomputed and reused across different parts of the calculation to minimize redundant operations.

3.3 Evaluation of the Spatial Matrix Element

The time-dependent matrix element $M_q(t)$ is a key ingredient in both the quantum and semiclassical formulations. It involves an integral over the radial coordinate and the polar angle, including oscillatory exponential terms and confluent hypergeometric functions.

The angular integration was performed using Gauss–Legendre quadrature, while the radial integration was evaluated on a uniform grid. A simplified version of the numerical implementation is shown below.

```

def compute_M_q_numeric(tr, omega_x_au, kL_tr, q0, Z,
                        N_r=150, R_max=150.0, N_mu=120):

    mu_nodes, mu_weights = leggauss(N_mu)
    r_grid = np.linspace(1e-6, R_max, N_r)

    norm_u0 = (Z**3/np.pi)**0.5
    u0_r = norm_u0 * np.exp(-Z * r_grid)
    q_eff = q0 + kL_tr
    v = Z/q_eff
    pref = np.exp(np.pi*v/2) * gamma(1.0 - 1j*v) / (2.0*np.pi)**1.5

    kx = omega_x_au/c_au

    integrand_r = np.zeros_like(r_grid, dtype=complex)

```

```

for ir, r in enumerate(r_grid):
    s = kx*r*np.sqrt(1.0 - mu_nodes**2)
    j0_vals = j0(s)
    z_arg = 1j*q_eff*r*(mu_nodes-1.0)
    # Most time consuming part of the simulation
    #confluent hypergeometric function
    oneF1 = np.array([hyp1f1_complex(-1j*v, 1.0, zz) for zz in z_arg],
dtype=complex)
    u_q_mu = pref*oneF1
    exp_factor = np.exp(1j*q_eff*r*mu_nodes)
    integrand_mu = mu_nodes*j0_vals*exp_factor*u_q_mu
    mu_integral = np.dot(integrand_mu, mu_weights)
    integrand_r[ir] = 2*np.pi*r**3*u0_r[ir]*mu_integral

val = np.trapezoid(integrand_r, r_grid)

return val

```

The use of high-precision evaluation for the confluent hypergeometric function was essential for numerical stability, particularly at higher photon energies.

3.4 Quantum-Mechanical Probability

The quantum-mechanical double differential probability requires a coherent time integral of the matrix element multiplied by a rapidly oscillating phase. The cumulative action appearing in the phase was precomputed to reduce numerical overhead. Additional parallelization and memory optimization techniques were also employed which is discussed further.

```

def DP_single_optimized(args):
    """Calculates DP for a single frequency using Shared Memory"""
    omega_x_eV, Nr, R_max, N_mu = args

    # Access shared arrays directly
    t_grid = shared_t_grid
    kL_t = shared_kL_t

    omega_x_au = omega_x_eV / hartree_ev
    Nt = len(t_grid)
    I0_au = -0.5 * (Z**2)

    =
    J = np.zeros(Nt, dtype=float)
    # Vectorized J calculation for speed
    integrand = (q0 + kL_t)**2
    J[1:] = np.cumsum(0.5 * (integrand[1:] + integrand[:-1]) * np.diff(t_grid))

```

```

integrand_arr = np.zeros(Nt, dtype=complex)
for i, tval in enumerate(t_grid):
    kL_tr = kL_t[i]
    M_tr = compute_M_q_numeric(tval, omega_x_au, kL_tr, q0, Z,
                               N_r=Nr, R_max=R_max, N_mu=N_mu)

    phase_factor = np.exp(1j * ((omega_x_au + I0_au) * tval) - 0.5j * J[i])
    integrand_arr[i] = phase_factor * M_tr

A = np.trapezoid(integrand_arr, t_grid)
pref = (omega_x_au**3) / ((2.0 * np.pi)**2 * c_au**3)
DP_val = pref * (np.abs(A)**2)

gc.collect()

return (omega_x_eV, DP_val)

```

Due to its computational cost, this calculation was restricted to selected parameter sets and used primarily for validation of the semiclassical model.

3.5 Semiclassical Probability and Root Finding

The semiclassical calculation requires identifying the times at which the instantaneous photon energy matches a target value. These times were obtained by locating sign changes in $\omega_X(t) - \omega_{\text{target}}$.

```

def find_time_roots(omega_t, t_grid, omega_target):
    f = omega_t - omega_target
    idx = np.where(np.sign(f[:-1]) * np.sign(f[1:]) <= 0)[0]
    roots = []
    for i in idx:
        t1, t2 = t_grid[i], t_grid[i+1]
        f1, f2 = f[i], f[i+1]
        roots.append(t1 - f1*(t2-t1)/(f2-f1))
    return roots

```

The resulting recombination times were then used to evaluate the classical differential probability as an incoherent sum. This approach proved to be numerically stable and efficient, enabling extensive parameter sweeps over the carrier-envelope phase.

3.6 Parallelization and Performance Considerations

To make the simulations tractable, parallelization was employed using Python's `multiprocessing` module. Independent photon energies were distributed across available CPU cores, and shared-memory arrays were used to minimize memory duplication.

Despite these optimizations, the quantum-mechanical calculation remains computationally demanding, whereas the semiclassical approach allows rapid exploration of parameter space. This distinction strongly influenced the structure of the results presented in the following chapter.

Chapter 4

Results and Discussion

4.1 Validation Against Literature

We first validated our implementation using parameters from Bivona et al. (2006): Ti:Sapphire laser at $I_L = 6 \times 10^{14}$ W/cm², electron energy $\varepsilon_q = 90$ eV, hydrogen target, 2-cycle pulse with $\varphi = -90^\circ$.

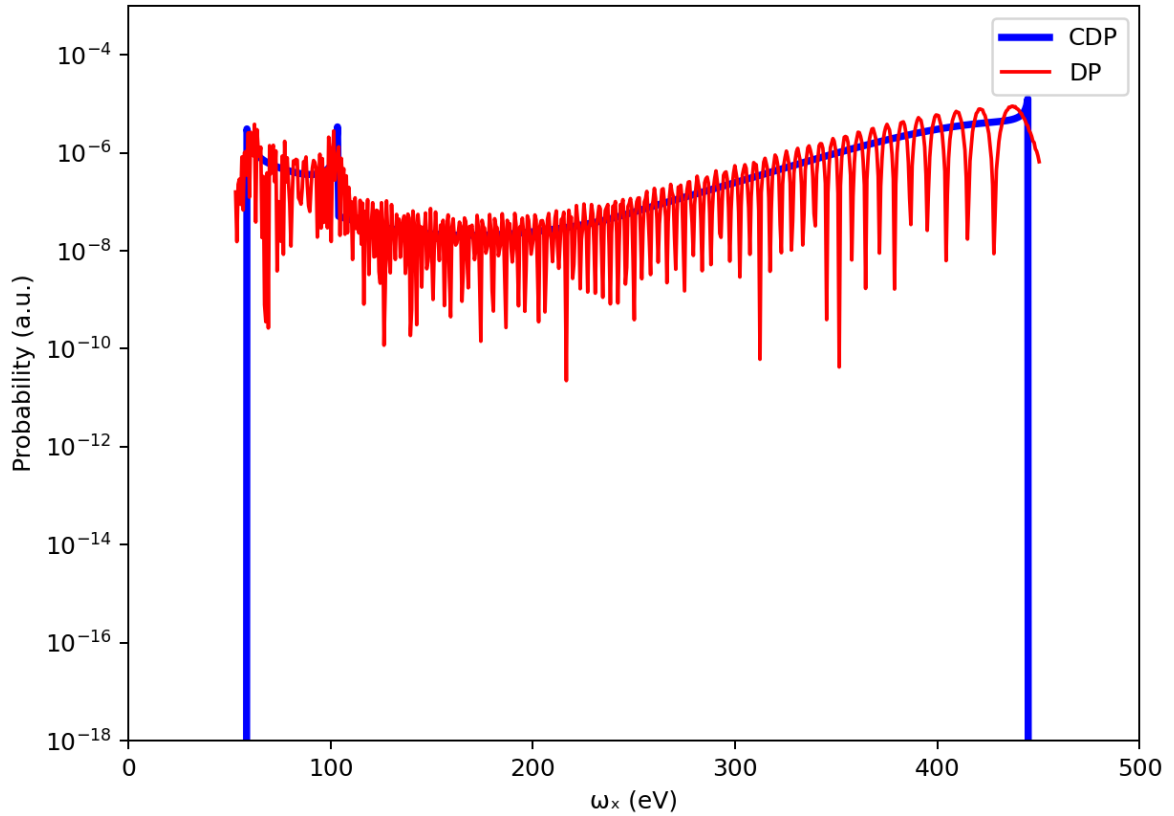


Figure 4.1: Comparison of Quantum (DP, red) and Semiclassical (CDP, blue) emission spectra. The quantum result shows rapid oscillations from interference, while the semiclassical result captures the envelope and cutoffs.

Figure 4.1 shows excellent agreement with published results. Key features include:

1. Emission is confined to roughly 50-450 eV, corresponding to the range of instantaneous kinetic energies during the pulse.
2. Prominent peaks appear at both cutoffs. The semiclassical model explains these as occurring when $E(t) \approx 0$ (electron turning points).
3. The DP curve exhibits rapid fringes from interference between different recombination times—absent in the CDP which sums probabilities incoherently.
4. Despite ignoring interference, the CDP correctly predicts cutoff positions and overall spectral shape, validating its use as a fast computational tool.

Although Fig. 4.1 confirms the correctness of the implementation for a single CEP value, it does not show how the spectrum changes as the laser waveform is varied. Since the semiclassical model efficiently captures the overall spectral features, it can be used to explore this dependence more systematically. Before performing a full CEP sweep, we therefore examine the CDP spectra for a few representative CEP values to visualize how the cutoff positions and spectral width evolve.

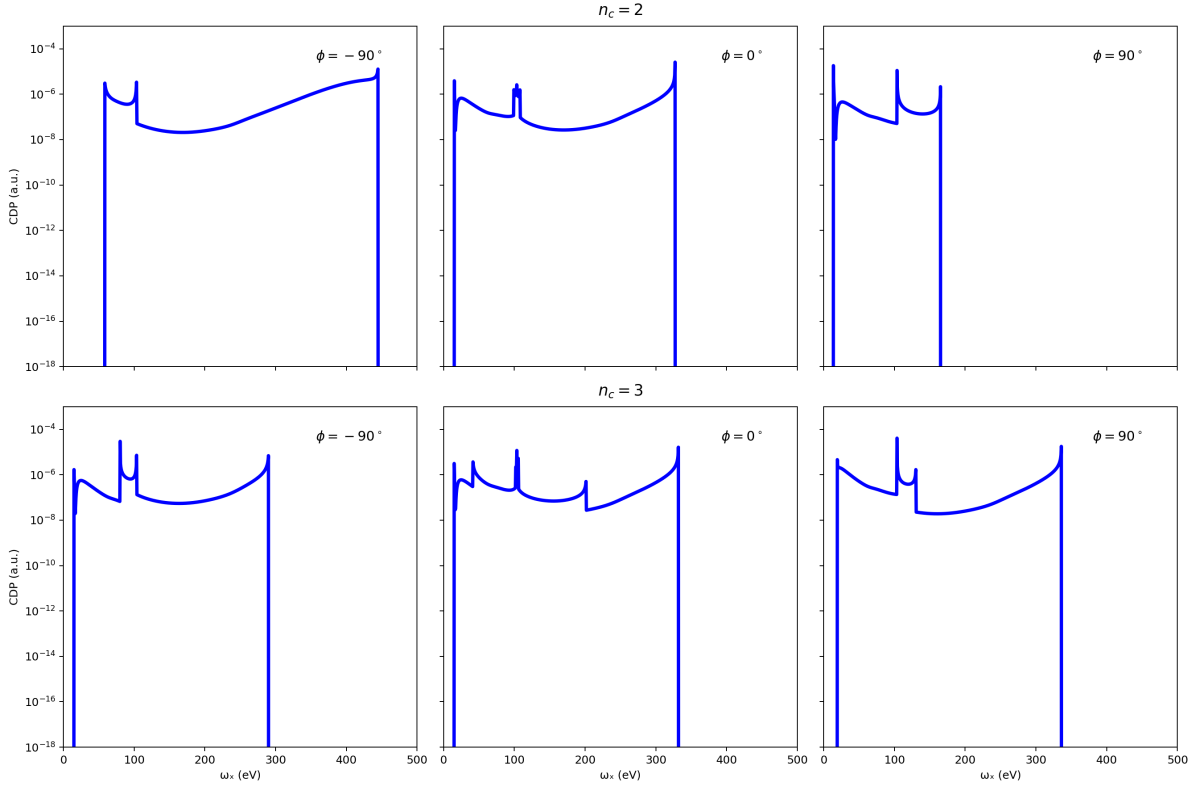


Figure 4.2: Semiclassical CDP spectra for a two-cycle and three cycle laser pulse ($n_c = 2, 3$) at three representative carrier-envelope phases ($\phi = -90^\circ, 0^\circ, 90^\circ$).

4.2 CEP Parameter Sweeps

We computed CDP for φ from -90° to $+90^\circ$ in 1° steps, for both $n_c = 2$ and $n_c = 3$.

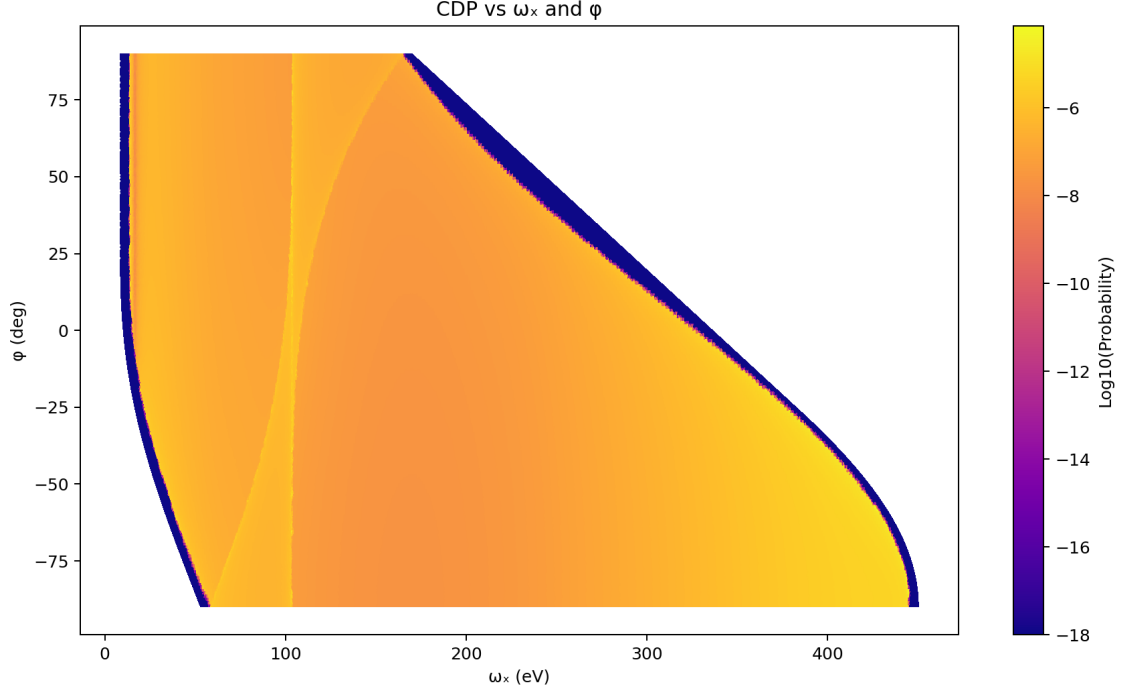


Figure 4.3: Semiclassical CDP heatmap showing the dependence of the emitted photon spectrum on the carrier-envelope phase for a two-cycle pulse ($n_c = 2$).

Figure 4.3 shows the CEP-dependent CDP heatmap for a two-cycle laser pulse. A strong modulation of the spectral cutoff energies with CEP is clearly visible. In particular, the maximum photon energy varies significantly as the CEP is changed, indicating a strong dependence of the electron's maximum instantaneous momentum on the laser waveform. The asymmetry of the pattern with respect to $\varphi = 0^\circ$ reflects the strongly waveform-dependent nature of few-cycle pulses, where shifting the carrier-envelope phase alters which half-cycle dominates the electron dynamics. The sharp boundaries of the allowed region correspond to classical cutoff energies and are enhanced due to turning points in the electron motion where $E(t) \approx 0$.

The corresponding heatmap for a three-cycle pulse is shown in Fig. 4.4. While the overall structure remains similar, the dependence of the cutoff energies on the CEP is noticeably reduced compared to the two-cycle case. The allowed spectral region varies more smoothly with φ , and the pattern exhibits increased symmetry. This behavior reflects the reduced sensitivity to CEP as the pulse duration increases, since the electron experiences several similar oscillations of the field and the effect of shifting the carrier phase is partially averaged out.

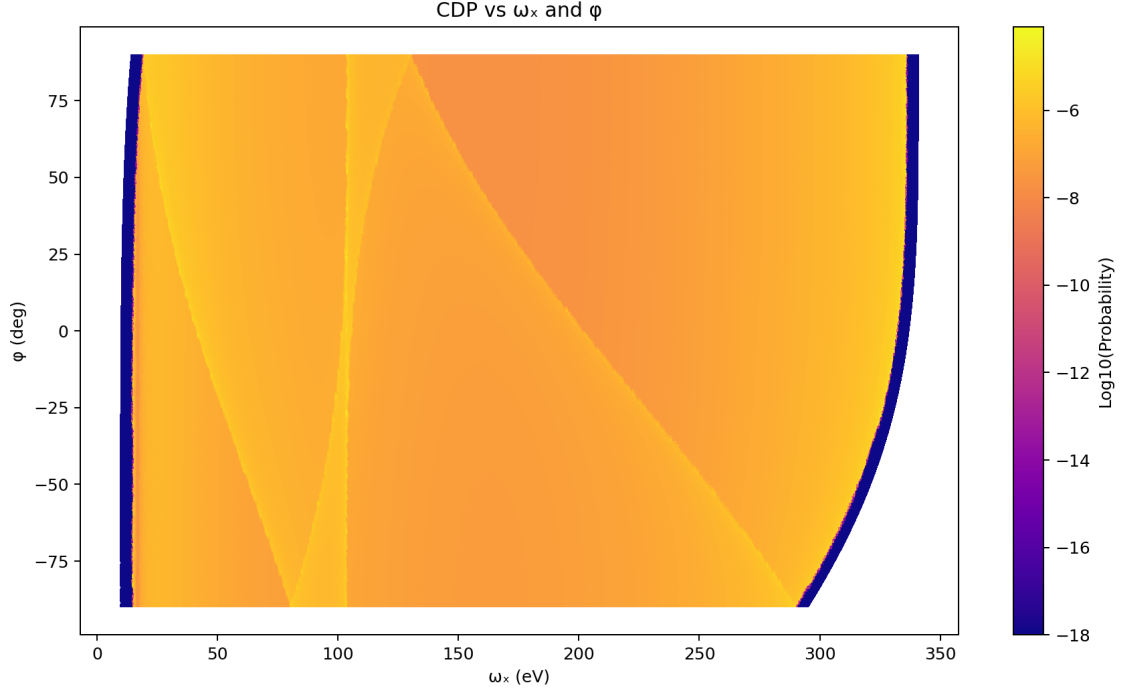


Figure 4.4: Semiclassical CDP heatmap showing the dependence of the emitted photon spectrum on the carrier-envelope phase for a three-cycle pulse ($n_c = 3$).

Together, Figs. 4.3 and 4.4 demonstrate that CEP-based control of laser-assisted radiative recombination is most effective in the ultrashort-pulse regime.

Chapter 5

Limitations and Future Directions

While the results presented in this project provide valuable insight into laser-assisted radiative recombination in the few-cycle regime, it is important to acknowledge the limitations inherent in the theoretical models and numerical implementation employed. Identifying these limitations not only clarifies the scope of the present work but also suggests directions for future investigation.

5.1 Validity of the Coulomb–Volkov Approximation

The quantum-mechanical description used throughout this work relies on the Coulomb–Volkov approximation for the continuum electron state. This approximation treats the interaction with the laser field exactly while incorporating the Coulomb potential of the ion in an approximate manner. Although this approach has been shown to provide reliable results for a wide range of strong-field processes, it neglects certain effects such as rescattering and higher-order Coulomb corrections.

In particular, the Coulomb–Volkov approximation becomes less accurate when the electron spends significant time near the ion core during the pulse, or when rescattering trajectories contribute appreciably to the recombination process. These effects are expected to be more relevant at lower electron energies or higher laser intensities than those considered in the present study.

5.2 Neglect of Electron–Electron and Plasma Effects

The present work considers recombination of a single electron with a hydrogenic ion in isolation. In realistic experimental or astrophysical environments, collective plasma effects, electron–electron interactions, and external fields may influence the recombination dynamics. Such effects are beyond the scope of the current model but could play an important role in dense or strongly coupled systems.

5.3 Numerical Constraints

From a computational perspective, the most significant limitation arises from the high numerical cost of the full quantum-mechanical calculation. The need to evaluate the spatial matrix element at every time step and for every photon energy makes extensive parameter scans impractical within reasonable computational time.

As a result, the quantum calculation was used primarily as a benchmark for selected parameter sets, while the semiclassical model was employed for broader explorations of carrier-envelope phase dependence. Although this strategy is well justified physically, it necessarily limits the resolution with which purely quantum interference effects can be explored across parameter space.

5.4 Possible Extensions

Several natural extensions of the present work can be envisioned. One immediate direction would be to explore the dependence of the recombination spectrum on laser intensity and polarization, allowing a more complete mapping of the control parameters available in LARR. Extending the analysis to ions with higher nuclear charge would also be of interest, as stronger Coulomb fields may modify both the validity of the semiclassical picture and the sensitivity to CEP.

From a theoretical standpoint, incorporating more accurate continuum wavefunctions or solving the time-dependent Schrödinger equation numerically would allow a more complete assessment of the approximations used here. Finally, extending the present analysis to include angularly resolved or polarization-resolved spectra could provide deeper insight into the underlying recombination dynamics.

Chapter 6

Conclusion

In this project, laser-assisted radiative recombination of a free electron with a hydrogenic ion in the presence of few-cycle laser pulses was investigated using both quantum-mechanical and semiclassical approaches. The primary objective was to understand how ultrashort laser pulses and the carrier-envelope phase influence the emission spectrum and to assess the extent to which simplified models can capture the essential physics of the process.

A numerical implementation of the full quantum-mechanical description based on the Coulomb–Volkov approximation was developed and used to reproduce key spectral features reported in the literature. In particular, the quantum calculations revealed pronounced interference structures arising from coherent contributions at multiple recombination times within the laser pulse. These oscillatory features highlight the intrinsically time-dependent and quantum nature of the recombination process.

In parallel, a semiclassical two-step model was implemented, offering a transparent physical interpretation of the emission spectra. Despite its simplicity, the semiclassical approach was shown to accurately predict the spectral cutoffs and overall envelope of the emission spectrum. This agreement confirms that the boundaries of the spectrum are governed primarily by classical electron dynamics in the laser field, while quantum interference effects modulate the spectrum within the classically allowed region.

A central result of this work is the demonstration of strong carrier-envelope phase control in the ultrashort-pulse regime. For two-cycle pulses, the spectral width of the emitted radiation was found to depend sensitively on the CEP, whereas this dependence is significantly reduced for three-cycle pulses. This behavior reflects the increasing temporal symmetry of longer pulses and underscores the importance of few-cycle fields for coherent control of atomic processes.

Overall, the results presented here are consistent with and extend previous theoretical studies of laser-assisted radiative recombination. Beyond reproducing known results, this project provides a detailed computational perspective on the interplay between quantum interference, classical dynamics, and ultrafast laser control. The experience gained

through implementing and analyzing both quantum and semiclassical models offers valuable insight into time-dependent strong-field physics and highlights the trade-offs between accuracy and computational efficiency in practical simulations.

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